T2

1) NPW = $-100{,}000 + 50{,}000\,(P|F, 8\%, 25)$ + $(25000-6000-8000)(P|A, 8\%, 25)$

.1460 10.6748

(c)

$= 27{,}722.8$

2) (b) $t=1$: $-2000(1.05) = -2100 - 1000 = -3100$

$-3100(1.05) + 800 = -2455$

$-2455(1.05) + 800 = -1777.75$

$-1777.75(1.05) + 800 = -1066.6375$

$t=5$: $-1066.6375(1.05) + 800 = -319.9694$

$t=6$. $-319.9694(1.05) + 800 = 465.0321$.

$$\frac{319.9694}{(319.9694 + 465.0321)} = \frac{X \text{ months}}{12 \text{ months}}.$$

$X = 4.89$ months. ≈ 5 months.

5 yrs. 5 months.

3) (d) $-30{,}000 + X\,(P|F, 12\%, 15) + (4000-2000)(P|A, 12\%, 15) \geq 0$

.1827 6.8109

$X \geq 89645.32$

4) (a) $A = 200 - [100\,(F|A, 12\%, 4) + 50\,(F|A, 12\%, 2)](A|F, 12\%, 6)$

4.7793 2.1200 .1232

$= 128.0598$

$CE = \frac{128.0598}{.12} = 1067.165$

5) (c) $CE = 800{,}000 + \frac{50K\,(A|F, 12\%, 15)}{.12} + 80{,}000\,(P|F, 12\%, 10) + 25000\,(P|A, 12\%, 12)$

.0268 .3220 6.1944

$+ \frac{40{,}000}{.12}\,(P|F, 12\%, 12) = 1{,}077{,}353.333$

.2567

6) (e) $AEC = 200{,}000\,(A|P, 8\%, 15) - 5000\,(A|F, 8\%, 15) + (40K + 12K + 2K)$

.1168 .0368

$= 77{,}176$

# of rides : $R \times .25 \geq 77{,}176$

$R \geq 308{,}704$

7) (d) $AE(i^*) = 700 + [400(P|F, i^*, 1) - 100(P|F, i^*, 3)](A|P, i^*, 4)$

$PW = -10000 + \frac{AE(i)}{i}$

$= -10000 + \frac{700 - [400(P|F, i^*, 1) + 100(P|F, i^*, 3)](A|P, i^*, 4)}{i^*}$

$PW(i^*) = 0$

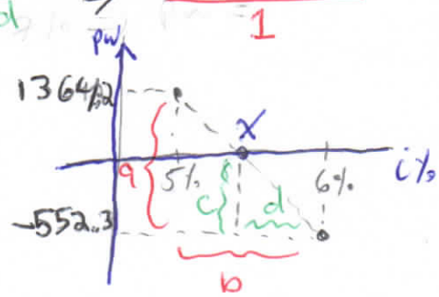
if you try 5% (if you try 4% $\Rightarrow (+) \Rightarrow$ Need to $\uparrow r$

$PW(5\%) = 1364.2024$ $\Rightarrow$ Need to $\downarrow$ weight on future (+)'s

$\Rightarrow \uparrow r \Rightarrow \downarrow \delta$

$\Rightarrow PW(6\%) = -552.2825$

$\frac{a}{b} = \frac{c}{d} \Rightarrow \frac{1916.4849}{1} = \frac{552.2825}{6 - x}$



$6 - x = .2882$

$\boxed{5.7118 = x}$

8) (d) Try the middle first : 11.89%

$PB_1 = -7000(1.1189) + 10000 = 2167.7$

$PB_2 = 2167.7(1.06) - 20000 = -17702.238$

$PB_3 = -17702.238(1.1189) + 20000 = 192.97 > 0$ (Need to $\uparrow r$ to $\downarrow PW$ $\Rightarrow$ decrease weight on future)

trying 12.63%

$PB_1 = -7000(1.1263) + 10000 = 2115.9$

$PB_2 = 2115.9(1.06) - 20000 = -17757.146$

$PB_3 = -17757.146(1.1263) + 20000 = .1265 \approx$ zero.

9) (a) $PW^A(12\%) = -80{,}200 - (10{,}000)\overset{3.6048}{(P|A,12\%,5)} + 24000\,\overset{.5674}{(P|F,12\%,5)}$

$= -102630.4$

$PW^B(12\%) = -51000 - 13000\,\overset{3.6048}{(P|A,12\%,5)}$

$= -97862.4$

$PW^{B-A} = 4768$

10) (e) $AEC^A(10\%) = 50{,}000 \times .1 + 20{,}000\,\overset{.1054}{(A|F,10\%,7)} = 7108$

$AEC^B(10\%) = 200{,}000 \times .1 + 110{,}000\,\overset{.0023}{(A|F,10\%,40)} = 20253$

$AEC^{B-A} = 13145$

11) (b) $AEC^A(12\%) = \$1m\,\overset{.1468}{(A|P,12\%,15)} - 50k\,\overset{.0268}{(A|F,12\%,15)} + 60k$

$= 205{,}460$

$AEC^B(12\%) = \$800{,}000\,\overset{.1339}{(A|P,12\%,20)} - 10k\,\overset{.0139}{(A|F,12\%,20)} + 75K$

$= 181{,}981$

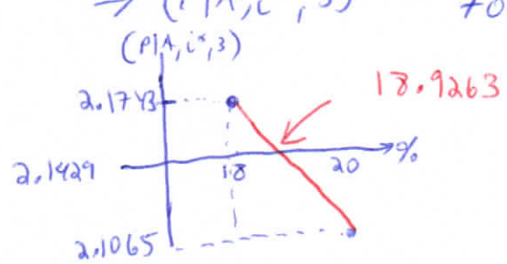
cost per tonne:

$C^A = \frac{205{,}460}{30 \times 365} = 18.76$ $\qquad C^A - C^B = \$2.14$

$C^B = \frac{181981}{30 \times 365} = \$16.62$

12) (d) $PW^A = -15000 + 7000(P|A, i^*, 3) = 0$

$\Rightarrow (P|A, i^*, 3) = \frac{15000}{7000} = 2.1429 \Rightarrow PW^A = 0.$



13) (c) A-B

1 : -500
2 : 1000
3 : -500

} non-simple investment. ⇒ PB approach

Letting $i^* = MARR = 10\%$

$t=1$: -500

$t=2$: $-500(1.1) + 1000 = 450$

$t=3$: $450(1.1) - 500 = -5$ ⇒ ↓ i^*

⇒ $i^* < MARR = 10\%$ ⇒ B wins.

(see example 6.4, step 3).

BONUS

15) (c)

n	A	B	B-A
0	-100	-200	-100
1	50	150	100
2	50	150-200=-50	-100
3	50-100=-50	150	200
4	50	150-200=50	-100
5	50	150	100
6	50	150	100

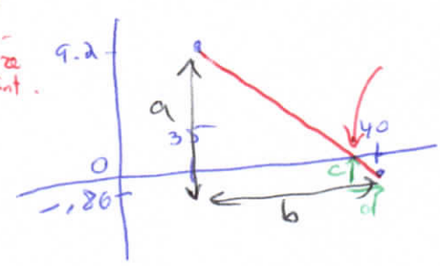
negative terms (+) pure incremental investment.

stays positive

$$PW = 0 = -100 + 100(P|A, i^*, 6) - 200\left[(P|F, i^*, 2) + (P|F, i^*, 4)\right] + 100(P|F, i^*, 3)$$

$PW(40\%) = -.86$ ⇐ should realize at this point.

$PW(35\%) = 9.2$



$$\frac{a}{b} = \frac{c}{d} \Rightarrow \frac{10.06}{5} = \frac{.86}{40-x}$$

$40 - x = .4274$

$39.57\% = x$ ⇐ TRY PB method ⇒ PB = 0

14) (d) $PW(10\%) = -100 + 100(P|A, 10\%, 6)$ (4.3553) $-200\left[(P|F, 10\%, 2)\right.$ (.8264) $+ (P|F, 10\%, 4)$ (.6830)$\left.\right]$ $+100(P|F, 10\%, 3)$ (.7513) $= 108.78$ of the LCM (6 yrs)

6 yrs ⇒ ANNUALIZING $AEW = 108.78(A|P, 10\%, 6) = 24.1759$ (.2296)